SymPy Bug Card

Bug 14: Rayleigh CDF is positive below support

Task. Compute the CDF of a Rayleigh random variable below its support.

$$R \sim \text{Rayleigh}(1), \qquad F_R(-1) = \text{cdf}(R)(-1)$$

SymPy's answer.

$$\text{cdf}(R)(-1) = 1 - e^{-1/2}.$$

Correct answer.

$$\text{cdf}(R)(-1) = 0.$$

Explanation. A Rayleigh random variable has support $[0, \infty)$, so the event $R \leq -1$ is empty. SymPy is extending the nonnegative-branch formula $1 - e^{-z^2/2}$ to $z < 0$, where the CDF should instead be zero.

Likely root cause. The diagnosis points to `sympy/stats/crv_types.py`: `RayleighDistribution._cdf` returns an unguarded formula even though the distribution declares support $[0, \infty)$, and `sympy/stats/crv.py` accepts that custom CDF before generic support handling. Deduplication frames this as a new failure mode with no public precedent.

Artifact (GitHub).
bug-report/bug-014-rayleigh-cdf-is-positive-below-support/